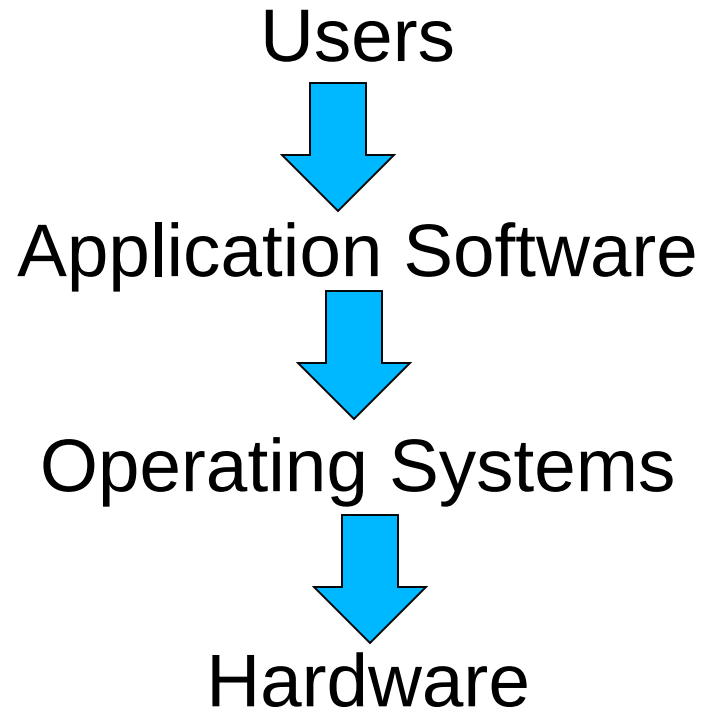


MA511:Computer Programming Introduction

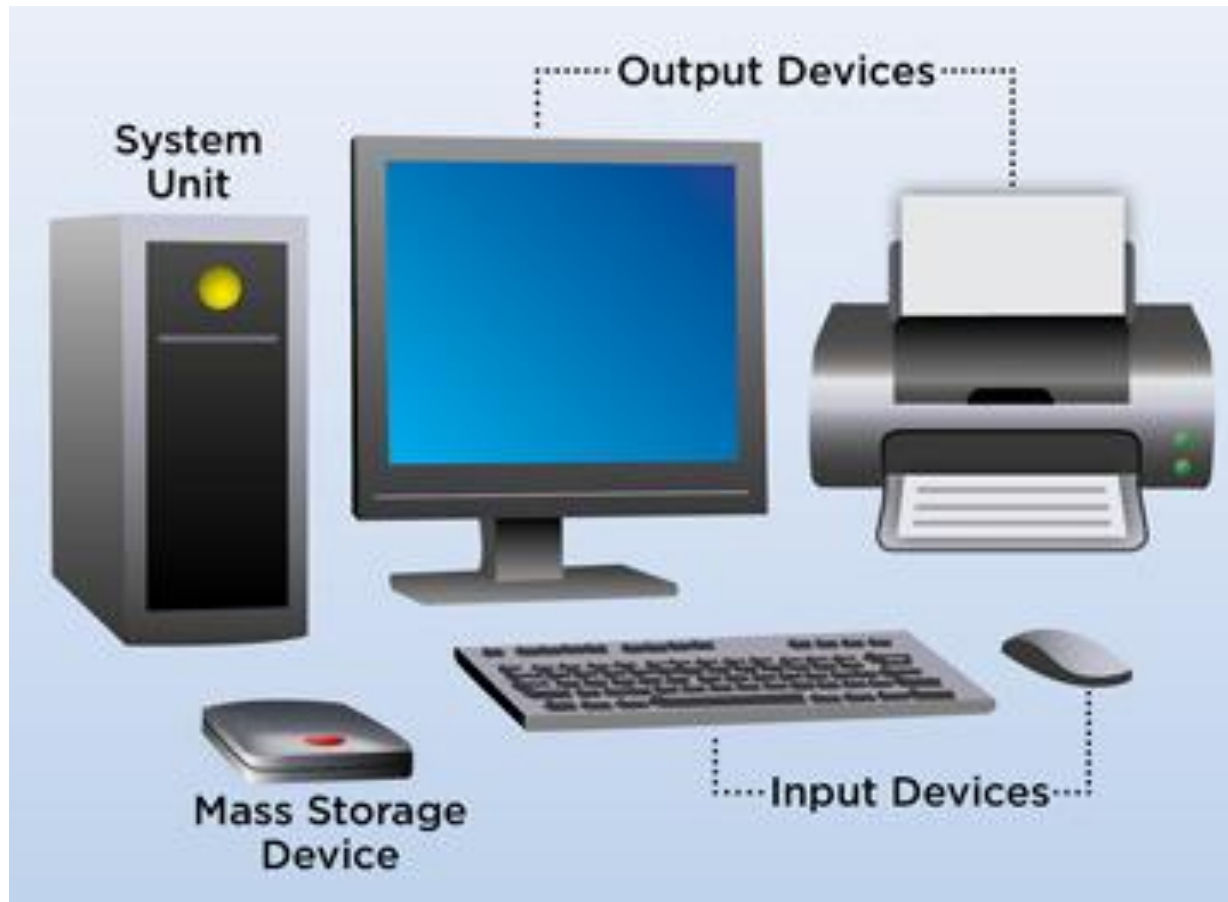
Instructor: Ashok Singh Sairam
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Computer Systems:



Hardware

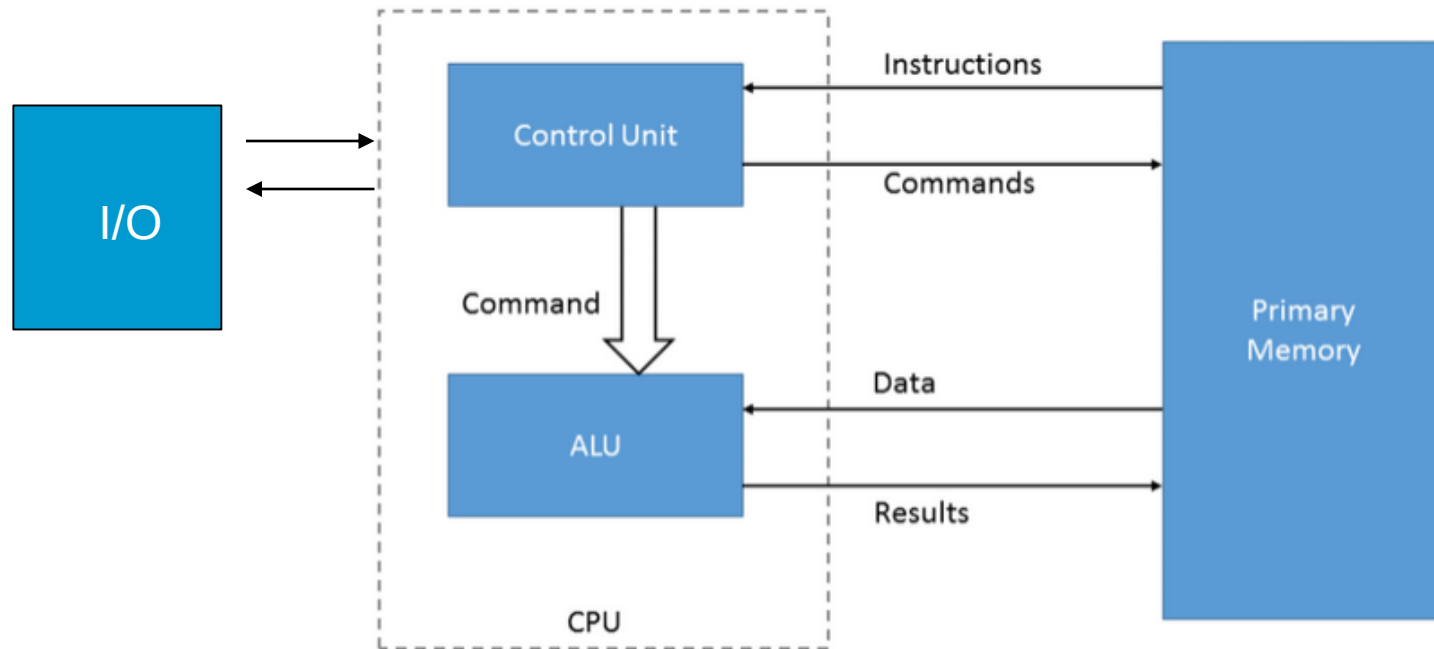


Inside system unit

- Common components: Power Supply, CPU, memory, HDD, adapter cards, ports, drive bays



Functions of System Unit



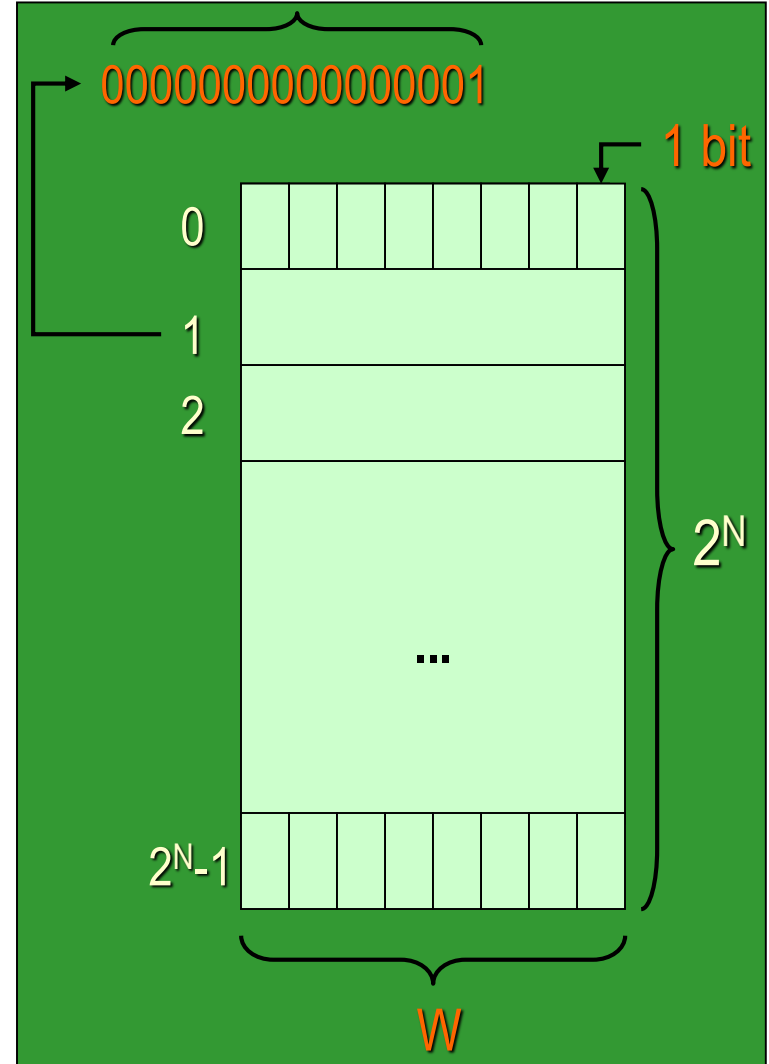


The CPU

- Components of CPU
 - Control Unit (CU), Arithmetic Logic Unit (ALU), Registers, Decode unit, etc.
- CU: Executes instructions
 - Fetch instructions, decode and execute
- ALU: perform the mathematical operations (+, -, x, /, ...) and logic operations (=, <, >, and, or, not)

Random Access Memory

- Need to distinguish between
 - the address of a memory cell and the content of a memory cell
- Memory width (W):
 - # bits that can be processed in parallel, Ex. 64-bit
- Address width (N):
 - #bits used to represent each address, determines the maximum memory size = address space
 - If address width is N -bits, then address space is 2^N ($0, 1, \dots, 2^N - 1$)





Input/Output Subsystem

- Handles devices that allow the computer system to:
 - Communicate and interact with the outside world
 - Screen, keyboard, printer, ...
 - Store information (mass-storage)
 - Hard-drives, floppies, CD, tapes, ...
- Mass-Storage Device Access Methods:
 - Direct Access Storage Devices (DASDs)
 - Hard-drives, floppy-disks, CD-ROMs, ...
 - Sequential Access Storage Devices (SASDs)
 - Tapes (for example, used as backup devices)



How computers manipulate data?

- Computers (digital devices) work with discrete data, digits 1 and 0, or like a switch – on or off
- These 1s and 0s are referred to as binary digits or shortened to bits
- Computers use sequences of bits to digitally represent numbers, letters, punctuation marks, music, pictures, and videos



Bit Basics

- Bit: From Binary digit
 - Smallest unit of information computer can process
 - Can have one of two values: 0 or 1
- Byte: Collection of 8 bits
 - Can represent 256 different messages ($256 = 2^8$)
- Bits as numbers

Decimal	Binary	Decimal	Binary
0	0000	5	0101
1	0001	6	0110
2	0010	7	0111
3	0011	8	1000
4	0100	9	1001



Bits as Codes

- Codes represent each letter, digit, and special character
- ASCII: Most widely used
 - Each character is a unique 8-bit code
 - 256 unique codes for 26 letters, 10 digits, special characters
- Unicode: Supports more than 100,000 unique characters

Character	ASCII binary code
A	0 1 0 0 0 0 0 1
B	0 1 0 0 0 0 1 0
C	0 1 0 0 0 0 1 1
D	0 1 0 0 0 1 0 0
E	0 1 0 0 0 1 0 1
F	0 1 0 0 0 1 1 0
G	0 1 0 0 0 1 1 1
H	0 1 0 0 1 0 0 0
I	0 1 0 0 1 0 0 1
J	0 1 0 0 1 0 1 0
K	0 1 0 0 1 0 1 1
L	0 1 0 0 1 1 0 0
M	0 1 0 0 1 1 0 1
N	0 1 0 0 1 1 1 0
O	0 1 0 0 1 1 1 1
P	0 1 0 1 0 0 0 0
Q	0 1 0 1 0 0 0 1
R	0 1 0 1 0 0 1 0
S	0 1 0 1 0 0 1 1
T	0 1 0 1 0 1 0 0
U	0 1 0 1 0 1 0 1
V	0 1 0 1 0 1 1 0
W	0 1 0 1 0 1 1 1
X	0 1 0 1 1 0 0 0
Y	0 1 0 1 1 0 0 1
Z	0 1 0 1 1 0 1 0
0	0 0 1 1 0 0 0 0
1	0 0 1 1 0 0 0 1
2	0 0 1 1 0 0 1 0
3	0 0 1 1 0 0 1 1
4	0 0 1 1 0 1 0 0
5	0 0 1 1 0 1 0 1
6	0 0 1 1 0 1 1 0
7	0 0 1 1 0 1 1 1
8	0 0 1 1 1 0 0 0
9	0 0 1 1 1 0 0 1



Memory Size / Speed

- Typical memory in a personal computer (PC):
 - in Gigabytes
- Memory sizes:
 - Kilobyte (KB) = 2^{10} = 1,024 bytes ~ 1 thousand
 - Megabyte (MB) = 2^{20} = 1,048,576 bytes ~ 1 million
 - Gigabyte (GB) = 2^{30} = 1,073,741,824 bytes ~ 1 billion
 - Terabyte (TB) = 2^{40} = 1,099,511,627,776 bytes ~ 1 trillion
- Memory Access Time (read from/ write to memory)
 - 50-75 nanoseconds (1 nsec. = 0.000000001 sec.)
- RAM is
 - volatile (can only store when power is on)
 - relatively expensive



Machine Language Instructions

- A machine language instruction consists of:
 - Operation code, telling which operation to perform
 - Address field(s), telling the memory addresses of the values on which the operation works.
- Ex: Add contents of memory locations X and Y, and store back result in memory location Y
- Assume: opcode for ADD is 9, and addresses X=99, Y=100

Opcode (8 bits)	Address 1 (16 bits)	Address 2 (16 bits)
00001001	0000000001100011	0000000001100100



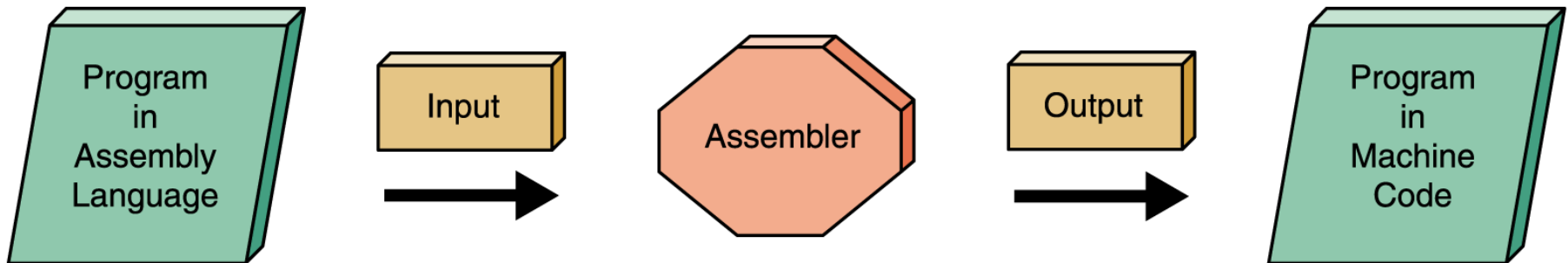
Assembly Language

- **Assembly languages:** assign **mnemonic** letter codes to each machine-language instruction
 - Same example: ADD X, Y: **Add contents of memory locations X and Y, and store back result in memory location Y**

Mnemonic	Operand	Meaning
LOADA	h#0063,d	Load the contents of location X into register A
ADDA	h#0064,d	Add contents of location Y into register A
STOREA	h#0064,d	Store the contents of register A into location Y

Assembler

- A program called an assembler reads each of the instructions in mnemonic form and translates it into the machine-language equivalent





High level language (HLLs)

- HLLs are simple languages that use English and mathematical symbols like +, -, %, / etc. for its program construction, eg. C, Fortran, Java etc.
 - easy to learn and use
 - has to be converted to machine language for the computer to understand - Compiler or Interpreter
 - **Compiler** is a program that translates the instruction of a higher level language to machine language.
 - An **interpreter** is also a translator but it takes one statement of higher level language, translate it into machine language and immediately execute it.



ALGORITHMS AND FLOWCHARTS

- Typical programming task can be divided into 2 phases:
- ***Problem solving phase***
 - produce an ordered sequence of steps that describe solution of problem
 - this sequence of steps is called an ***algorithm*** and the code is called pseudocode
 - graphical representation is called ***flowchart***
- ***Implementation phase***
 - implement the program in some programming language



Algorithm

- Precise step-by-step instructions to solve a problem.
- Characteristics
 - Accept Inputs.
 - Instructions are precise and unambiguous.
 - Instructions are simple.
 - Takes finite time.
 - Produces one or more output.



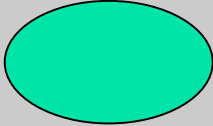
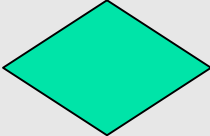
The Flowchart

A graphical representation of the sequence of operations in an information system or program.

A Flowchart

- shows logic of an algorithm
- emphasizes individual steps and their interconnections
- e.g. control flow from one action to the next

Flowchart Symbols

Symbol	Shape	Meaning
Oval		Start/End
Parallelogram		Input/Output
Rectangle		Processing
Diamond		Decision
Arrow		Connector



Example:

- Write an algorithm/flowchart that reads two values, determines the largest value and stores it in a third variable.

Print the largest value, with an identifying message.



Review: Exercise

- INPUT VALUE1, VALUE2, VALUE3
- NUM $\leftarrow$ 10
- IF (VALUE1 > VALUE2) THEN
- NUM $\leftarrow$ VALUE1
- ELSE
- VAR $\leftarrow$ 40
- IF (NUM < VALUE3)
- VAR $\leftarrow$ 30
- NUM $\leftarrow$ VAR - NUM
- PRINT NUM
- If VALUE1 = 20, VALUE2 = 30, and VALUE3 = 10, what will be the algorithm print?



Course Information:

- Course Page: <http://www.iitg.ac.in/ashok/MA511>
- Textbook: A. Kelley and I. Pohl, *A Book on C*, 4th Ed
- Reference:
 - P.J.Deitel & H.M.Deitel , *C How To Program*, 5th Ed
 - Kernighan & Ritchie, *The C Programming Language*, 2nd Ed.
- Syllabus:
 - Computer Fundamentals,
 - C Programming (*for details check course website*)



Course Evaluation

- Mark distribution
 - Class Performance(Quiz, etc.) : 10%
 - Lab Assignment/Exam : 20%
 - Mid-Sem : 30%
 - End-Sem : 40%
- No make-up exam for quiz/mid-semester if you fail to appear in the exam.
- Attendance compulsory in all theory and lab classes



Class Timings

	8-8:55	9-9:55	10-10:55	11-11:55	12-12:55		2 – 4
Mon							
Tue	Lec						
Wed		Lec					
Thu			Lec				
Fri							Lab

Class Room

Theory: Rm 1G1,

Lab : Dept. Computer Lab, E1